HOMEWORK 1.4 · UNIT 1 · NC.8.NS.1 · NC.8.NS.2 · NC.8.EE.2

# Converting Numbers; Irrational Numbers

Aug 13 / Aug 14 · Targets: NS.1-1, NS.2-2, EE.2-2, NS.1-3, NS.1-5, NS.2-5, NS.2-8, EE.2-7

Name _____ Date _____ Block _____

**Remember: 9's rule:** the repeating digits go on top; one 9 for each repeating digit on the bottom. $0.\overline{27} = \frac{27}{99} = \frac{3}{11}$. **Estimate a root:** trap it between perfect squares. $36 < 40 < 49$, so $\sqrt{40}$ is between 6 and 7. **4-step algorithm:** $\sqrt{40} \approx 6 + \frac{40 - 36}{49 - 36} = 6 + \frac{4}{13} \approx 6.3$

**SCORE 3.0** Approaching · the basics

Targets: NS.1-1, NS.2-2, EE.2-2, NS.1-3

NS.1-1

**1** Write $\frac{5}{8}$ as a decimal.

NS.2-2

**2** $\sqrt{75}$ is between which two whole numbers?

(A) 7 and 8
(B) 8 and 9
(C) 9 and 10
(D) 37 and 38

NS.2-2

**3** Use a calculator. Round $\sqrt{33}$ to the nearest tenth.

REVIEW 1.3 EE.2-2

**4** $\sqrt[3]{125} = 5$

True False

REVIEW 1.2 NS.1-3

**5** Circle one: $0.636363...$

Rational Irrational

Why? _____

**SCORE 4.0** Proficient · on target

Targets: NS.1-5, NS.2-5

**6** Use the 9's rule. Write $0.\overline{5}$ as a fraction. NS.1-5

**7** Write $0.\overline{18}$ as a fraction in simplest form. NS.1-5

**8** **Jamal** made a mistake. Circle it. Explain it. Fix it. NS.2-5

JAMAL'S WORK

Estimate $\sqrt{30}$:

$5 + \frac{30 - 25}{30} = 5 + \frac{5}{30} \approx 5.2$

**SCORE 5.0** Mastery · go deeper

Targets: NS.2-8, EE.2-7

**9** The algorithm gives $\sqrt{83} \approx 9.1$. Prove it by squaring 9.1 and 9.2. NS.2-8

$9.1^2$ = _____ and $9.2^2$ = _____ , so _____ .

**10** Evaluate. Show each step. $\sqrt{49} - \sqrt[3]{8} + 3^2$ REVIEW 1.3 EE.2-7